

(d) (i) It is suggested that the quantities T_Q and T_P are related by the equation

$$T_Q = A + BT_P$$

where A and B are constants.

Using your answers in (c)(iii), determine the values of A and B .
Give appropriate units.

$A =$

$B =$

[2]

(ii) Theory suggests that

$$R = F \sin \theta - \frac{2A}{S}$$

where R is the weight of the rod and F has the value 2.94 N.

Use your answers in (a)(i) and (d)(i) to determine a value for R .
Give an appropriate unit.

$R =$ [1]

[Total: 20]

